

ECE 3441 Spring 2021 Exam 2

- ? Open book, open notes
- ? The use of homework help sites such as Chegg, Course Hero, etc. are prohibited.
- ? Collaboration between students is not allowed.
- ? 80 minutes

Submit a handwritten copy of the following statement to UH Blackboard:

This is my own work; I did not browse for solutions or collaborate.

First Name, Last Name

Peoplesoft ID

Date DD/MM/YYYY

Question	Points	Maximum
1		24
2		24
3		24
4		24
5		24

Total: _____/100 possible points + 20 Bonus

Question #1: You must show work to receive credit.

Version A: (use the **3rd** digit of your PSID) (2, 4, 6, 7, 9)

Reduce the following state table to a minimum number of states. (5 states minimum)

Present State	Next State		Present Output (Z)
	X = 0	X = 1	
P	T	T	1
Q	R	T	1
R	X	W	0
S	W	P	1
T	X	U	0
U	T	V	0
V	W	Q	1
W	R	S	0
X	U	Q	1

Question #1: You must show work to receive credit.

Version B: (use the **3rd** digit of your PSID) **(0, 1, 3, 5, 8)**

Reduce the following state table to a minimum number of states. (5 states minimum)

Present State	Next State		Present Output (Z)
	X = 0	X = 1	
S0	S4	S4	0
S1	S2	S4	0
S2	S8	S7	1
S3	S7	S0	0
S4	S8	S5	1
S5	S4	S6	1
S6	S7	S1	0
S7	S2	S3	1
S8	S5	S1	0

Question #2: You must show work to receive credit.

Version A: (use the **4th** digit of your PSID) (5, 6, 7, 8, 9)

Given a counter which counts in the following sequence:

101, 001, 011, 111, 110, 000, 010, (repeat) 101, ...

Derive and simplify the input equations for this 3-bit counter using T flip-flops.

Question #2: You must show work to receive credit.

Version B: (use the **4th** digit of your PSID) (0, 1, 2, 3, 4)

Given a counter which counts in the following sequence:

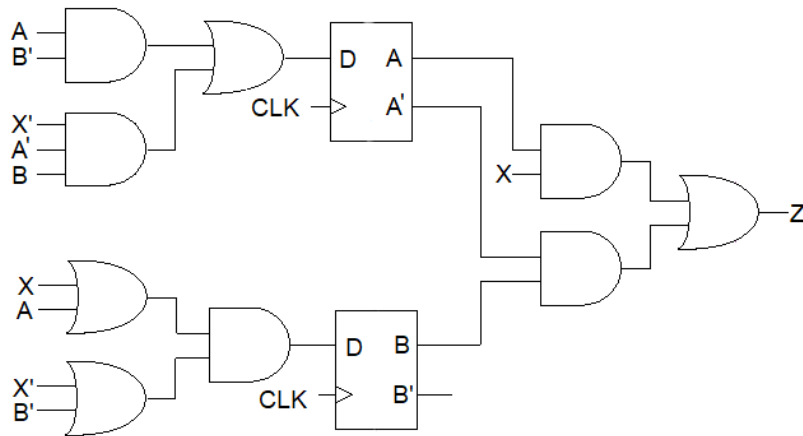
111, 011, 001, 101, 010, 000, 110, (repeat) 111, ...

Derive and simplify the input equations for this 3-bit counter using T flip-flops.

Question #3: You must show work to receive credit.

Version A: (use the **5th** digit of your PSID) (4, 5, 6, 7, 8)

Given a sequential circuit with input X and output Z:



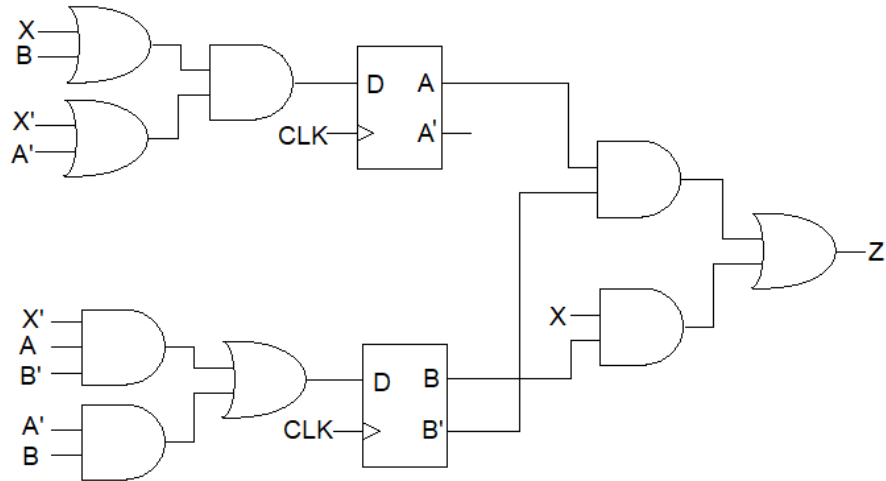
- a. Write the next-state equation for A and B.

- b. Construct the transition table for this circuit.

Question #3: You must show work to receive credit.

Version B: (use the **5th** digit of your PSID) (0, 1, 2, 3, 9)

Given a sequential circuit with input X and output Z:



- a. Write the next-state equation for A and B.

- b. Construct the transition table for this circuit.

Question #4: You must show work to receive credit.

Version A: (use the 6th digit of your PSID) (1, 3, 5, 7, 9)

A Moore sequential circuit has one input (X) and two outputs (Z_1 & Z_2). The output:

- 1) $Z_1 = 1$ if total number of 1's received is divisible by 3, otherwise $Z_1 = 0$.
- 2) $Z_2 = 1$ if total number of 1's received is divisible by 4, otherwise $Z_2 = 0$.

Example sequence:

X	0	0	1	1	0	0	1	1	0	1	1	0	0	0	0	1	1	0	1	0	1	1	0	1	1	1	0
Z ₁ (1)	1	1	0	0	0	0	1	0	0	0	1	1	1	1	1	0	0	0	1	1	0	0	0	1	0	0	0
Z ₂ (1)	1	1	0	0	0	0	0	1	1	0	0	0	0	0	0	0	1	1	0	0	0	0	0	1	0	0	0

Note: 0 is divisible by all numbers.

Derive a state graph for the Moore sequential circuit. Make sure that the graph is neat and clearly labeled. (**12 states minimum**)

Question #4: You must show work to receive credit.**Version B:** (use the **6th** digit of your PSID) (0, 2, 4, 6, 8)A Moore sequential circuit has one input (X) and two outputs (Z_1 & Z_2). The output:

- 1) $Z_1 = 0$ if total number of 0's received is divisible by 3, otherwise $Z_1 = 1$.
- 2) $Z_2 = 1$ if total number of 0's received is divisible by 4, otherwise $Z_2 = 0$.

Example sequence:

X	1	1	0	0	1	1	0	0	1	0	0	1	1	1	1	0	0	1	0	1	0	0	1	0	0	0	1
Z₁ (0)	0	0	1	1	1	1	0	1	1	1	0	0	0	0	0	1	1	1	0	0	1	1	1	0	1	1	1
Z₂ (1)	1	1	0	0	0	0	0	1	1	0	0	0	0	0	0	0	1	1	0	0	0	0	0	1	0	0	0

Note: 0 is divisible by all numbers.Derive a state graph for the Moore sequential circuit. Make sure that the graph is neat and clearly labeled. (**12 states minimum**)

Question #5: You must show work to receive credit.

Version A: (use the **last digit** of your PSID) (1, 4, 5, 7, 9)

A Mealy sequential circuit has one input (X) and one output (Z). The output, $Z = 1$ if every **fourth** bit is the same as the bit from **three** cycles ago, otherwise $Z = 0$.

Note: The fourth bit is the first bit to the next four bits.

Example sequence:

X:	1	0	0	1	0	0	1	0	0	0	1	1	0	1	0	1
Z:	0	0	0	1	0	0	1	0	0	0	0	0	1	0	0	0

Derive a state graph of the Mealy sequential circuit. Make sure that the graph is neat and clearly labeled. (**7 minimum states**)

Question #5: You must show work to receive credit.

Version B: (use the **last digit** of your PSID) (0, 2, 3, 6, 8)

A Mealy sequential circuit has one input (X) and one output (Z). The output, $Z = 0$ if every **fourth** bit is **NOT** the same as the bit from **three** cycles ago, otherwise $Z = 1$.

Note: The fourth bit is the first bit to the next four bits.

Example sequence:

X:	0	0	0	1	0	0	0	0	0	0	1	1	0	1	0	1
Z:	1	1	1	0	1	1	0	1	1	1	1	1	1	1	1	0

Derive a state graph of the Mealy sequential circuit. Make sure that the graph is neat and clearly labeled. (**7 minimum states**)